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Roll no.

MID Semester of 3rd Sem B.Tech.

MID – SEMESTER EXAMINATION, SEPTEMBER, 2023

COURSE CODE: EAMTC04

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 25

Note:- Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	Examine if the set $W = \{(a_1, a_2, 1) : a_1, a_2 \in R\}$ is a subspace of R^3 .	2	CO1
1b	Check whether $S = \{(1,2,3), (1,1,1), (1,0,1)\}$ of R^3 is linearly independent or linearly dependent	3	CO1
2a	Let $S = \{(1,0), (1,1)\}$. Is S basis of R^2 .	2	CO1
2b	$T: R^2 \rightarrow R^2$ defined by $T(x, y) = (2x + y, x)$. Check if T is a linear transformation.	3	CO1
3a	$T: R^3 \rightarrow R^2$ is a linear transformation, defined by $T(x, y, z) = (x - y, 2z)$. Kernel and Image of T.	4	CO1
3b	Let $u = (2,3,2,1)$ and $v = (3,2,1,3)$ be two vectors of R^4 . Find L_2 norm of vector u, v .	1	CO2
4a	Use Gram-Schmidt process to convert the basis $\{(1, 2, 3), (1, 1, 1), (1, 0, 1)\}$ of the Euclidean space R^3 with standard inner product into an orthogonal basis.	4	CO2
4b	Examine whether set $\left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)\right\}$ is orthonormal set.	1	CO2
5a	Find Eigen values of the matrix $A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{bmatrix}$	2	CO2
5b	Show that the quadratic form $2x^2 + 2y^2 + 2z^2 + 2yz$ is positive definite.	3	CO2